



WEST BENGAL STATE UNIVERSITY

B.Sc. Major 2nd Semester Examination, 2025

CMSDSC202P-COMPUTER SCIENCE (MAJOR)

DATA STRUCTURE USING C++

PRACTICAL

SET-I

Time Allotted: 2 Hours

Full Marks: 25

The figures in the margin indicate full marks.

Candidates should answer in their own words and adhere to the word limit as practicable.

All symbols are of usual significance.

Answer any one question from the following

1. Write a program using recursion:
 - (i) to find the sum of first n natural numbers.
 - (ii) to compute the factors of a given number.
2. Write a program to implement Double Linked List. Include functions for insertion, deletion and display.
3. Write a program to search an element from a list. Give user the option to perform Binary search.
4. Write a program to sort a list of elements. Give user the option to perform sorting using Selection sort.
5. Write a program to implement Quick Sort.
6. Write a program to perform Queues operations using Circular Array implementation.
7. Write a program to implement Linked List and concatenate two linked lists.
8. Write a program to implement Circular Linked List. Include functions for insertion, display and reverse the list.
9. Write a program to implement Double Linked List. Include functions for insertion, display and reverse the list.
10. Write a program to scan a polynomial using linked list and add two polynomial.
11. Write a program to implement Diagonal Matrix using one-dimensional array. Write a program to add two diagonal matrices.
12. Write a program to create a Binary Search Tree and include following operations in tree:
 - (i) Insertion
 - (ii) Display its in-order traversals Recursively
 - (iii) Display height of tree.

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Answer any *one* question from the following

1. Write a program using recursion to calculate GCD of two numbers.
2. Write a program to search an element from a list. Give user the option to perform Binary search. Use Template functions.
3. Write a program to sort a list of elements. Give user the option to perform sorting using Selection sort.
4. Write a program to sort a list of elements. Give user the option to perform sorting using Insertion sort. Use Template functions.
5. Write a program to perform Stack operations using Array implementation. Test for stack overflow and underflow.
6. Write a program to implement Linked List. Include functions for insertion, display and reverse the list.
7. Write a program to implement Circular Linked List Include functions for insertion, deletion and display.
8. Write a program to scan a polynomial using Array and add two polynomial.
9. Write a program to implement Lower Triangular Matrix using a one-dimensional array. Add two Lower Triangular matrices.
10. Write a program to create a Binary Search Tree and include following operations in tree:
 - (i) Insertion
 - (ii) Display its in-order traversals Recursively
 - (iii) Count the non-leaf nodes and leaf nodes
11. Write a program to implement Stack using queues.
12. Write a program to implement heap sort.

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